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Activation of 5-HT_{1B} receptors inhibits the vasodepressor sensory CGRPergic outflow in pithed rats

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ABSTRACT

The importance of calcitonin gene-related peptide (CGRP) in the regulation of vascular tone has been widely documented. Indeed, stimulation of the perivascular sensory outflow in pithed rats results in vasodepressor responses, which are mediated by CGRP release. These vasodepressor responses are inhibited by clonidine via prejunctional $\alpha_{2A/2C}$-adrenoceptors, but no study has yet reported the role of prejunctional 5-hydroxytryptamine (5-HT) receptors in this experimental model. Since activation of prejunctional 5-HT_1 receptors results in inhibition of neurotransmitter release, this study sets out to investigate as an initial approach the role of 5-HT_{1B} receptors in the inhibition of the vasodepressor sensory outflow in pithed rats. Male Wistar pithed rats were pretreated with hexamethonium (2 mg/kg.min) followed by i.v. continuous infusions of methoxamine (20 μg/kg.min), and then by saline (0.02 ml/min) or CP-93,129 (a rodent 5-HT_{1B} receptor agonist; 0.1, 1 and 10 μg/kg.min). Under these conditions, electrical stimulation (0.56-5.6 Hz; 50 V and 2 ms) of the spinal cord (T_9-T_{12}) resulted in frequency-dependent decreases in diastolic blood pressure. The infusions of CP-93,129, as compared to those of saline, inhibited the vasodepressor responses induced by electrical stimulation without affecting those to i.v. bolus injections of exogenous α-CGRP (0.1, 0.18, 0.31, 0.56 and 1 μg/kg). This inhibition by CP-93,129 was abolished by the antagonists GR127935 (5-$HT_{1B/1D}$) or SB224289 (5-HT_{1B}), but not by BRL15572 (5-HT_{1D}). The above results suggest that CP-93,129-induced inhibition of the vasodepressor (perivascular) sensory outflow in pithed rats is mainly mediated by activation of prejunctional 5-HT_{1B} receptors.

1. Introduction

Resistance blood vessels are innervated by sympathetic and primary sensory nerves which play an important role in the regulation of vascular tone and in the maintenance of arterial blood pressure (Bell and McDermott, 1996; Rubino and Burnstock, 1996; Rubino et al., 1997). C fibres are primary sensory nerves originating from the spinal cord (Julius and Basbaum, 2001) and, upon stimulation, produce vasodilatation *via* the release of vasodilator neurotransmitters, particularly calcitonin gene-related peptide (CGRP; Kawasaki et al., 1988; Han et al., 1990a,b). In agreement with these findings, electrical stimulation of the perivascular sensory outflow in pithed rats results in vasodepressor responses (decreases in diastolic blood pressure) which are selectively blocked by $CGRP_{(8-37)}$, a CGRP receptor antagonist (Taguchi et al., 1992).

More recently, our group has shown that the above vasodepressor sensory outflow can be inhibited by clonidine via activation of prejunctional $\alpha_{2A/2C}$-adrenoceptors (Villalón et al., 2008), but little is known about the role of prejunctional 5-hydroxytryptamine (5-HT) receptors on CGRPergic vascular neurotransmission in this experimental model. Overall, activation of prejunctional 5-HT_1 receptors inhibits sympathetic neurotransmission in a wide variety of blood vessels and other organs (Fink and Göthert, 2007; Villalón and Centurión, 2007), but only one study using in situ hybridization techniques in rat dorsal root ganglion (DRG) has shown the expression of 5-HT_{1B} (but not 5-HT_{1A} and 5-HT_{1D}) receptors in small diameter neurons (Nicholson et al., 2003).

On this basis, considering that the 5-HT_1 receptor family consists of at least five subtypes, namely 5-HT_{1A}, 5-HT_{1B}, 5-HT_{1D}, 5-HT_{1E} and 5-HT_{1F} (see Hoyer et al., 1994; Villalón and Centurión, 2007), the present study in pithed rats was designed to investigate as an initial approach: (a) whether CP-93,129 (a rodent 5-HT_{1B} receptor agonist;

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Table 1
Binding affinity constants (pK_i) of CP-93,129 and several antagonists for 5-HT_1 receptor subtypes.

	5-HT_{1A}	5-HT_{1B}	5-HT_{1B} (rat)	5-HT_{1D}	5-ht_{1E}	5-HT_{1F}
CP-93,129	5.5[a] (rat)	6.4[b]	8.1[b]	5.7[a] (rat)	–	–
GR127935	7.2[c]	9.0[c]	8.8[b]	8.6[c]	5.4[c]	6.4[c]
SB224289	5.5[d]	8.0[d]	–	6.2[d]	<5.0[d]	<5.0[d]
BRL15572	7.7[c]	6.1[c]	–	7.9[c]	5.2[c]	6.0[c]

All data are given as pK_i values at human receptors, except when stated otherwise.
[a] Macor et al., 1990.
[b] Beer et al., 1998.
[c] Price et al., 1997.
[d] Hagan et al., 1997.

see Table 1) is capable of inhibiting the vasodepressor responses induced by either electrical stimulation of the perivascular sensory outflow or i.v. bolus injections of exogenous α-CGRP; and (b) the role 5-HT_{1B} receptors in this CP-93,129-induced inhibition by analyzing the effects of the antagonists GR127935 (5-$HT_{1B/1D}$), SB224289 (5-HT_{1B}) and BRL15572 (5-HT_{1D}) (see Table 1). Our results suggest that CP-93,129 inhibits the vasodepressor sensory outflow in pithed rats mainly by activation of prejunctional 5-HT_{1B} receptors.

2. Materials and methods

2.1. Animals

All animal procedures and the protocols of the present investigation were approved by our Institutional Ethics Committee on the use of animals in scientific experiments, according to the Guide for the Care and Use of Laboratory Animals in U.S.A. Male Wistar normotensive rats (300-350 g) were maintained at a12/12-h light/dark cycle (with light beginning at 07:00 h) and housed in a special room at constant temperature (22 ± 2 °C), and humidity (50 %), with food and water freely available in their home cages.

2.2. General methods

Experiments were carried out in a total of 90 rats. After anaesthesia with ether and cannulation of the trachea, the rats were pithed by inserting a stainless steel rod through the orbit and foramen magnum into the vertebral foramen (Gillespie et al., 1970). Then, the animals were artificially ventilated with room air using a model 7025 Ugo Basile pump (56 strokes per min; stroke volume = 20 ml/kg), as established by Kleinman and Radford (1964). After bilateral vagotomy, catheters were placed in: (i) the left and right femoral and jugular veins, for the continuous infusions of agonists (methoxamine and/or CP-93,129), and i.v. administration of antagonists, respectively; and (ii) the left carotid artery, connected to a Grass pressure transducer (P23XL), for the recording of arterial blood pressure. Heart rate was measured with a tachograph (7P4, Grass Instrument Co., Quincy, MA, U.S.A.), triggered from the blood pressure signal. Both blood pressure and heart rate were recorded simultaneously by a model 7 Grass polygraph (Grass Instrument Co., Quincy, MA, U.S.A.). At this point, the 90 rats were initially divided into two main sets, so that the effects produced by the continuous infusions of methoxamine and/or CP-93,129 under different treatments could be investigated on the vasodepressor responses induced by: (i) electrical stimulation of the perivascular (vasodepressor) sensory outflow (set 1; n = 75); or (ii) i.v. bolus injections of exogenous α-CGRP (set 2; n = 15). The vasodepressor stimulus–response curves and dose–response curves elicited by electrical stimulation and exogenous α-CGRP, respectively, were completed in about 50 min, and each response was elicited under unaltered values of resting blood pressure. The electrical stimuli (0.56, 1, 1.8, 3.1 and 5.6 Hz), as well as the dosing with α-CGRP (0.1, 0.18, 0.31, 0.56 and 1 µg/kg), were given using a sequential schedule at 5-10 min intervals (see below), as previously reported (Villalón et al., 2008). The body temperature of each pithed rat was maintained at 37 °C by a lamp and monitored with a rectal thermometer.

2.3. Experimental protocols

After the animals (n = 90) had been in a stable haemodynamic condition for at least 15 min, baseline values of diastolic blood pressure (a more accurate indicator of peripheral vascular resistance) and heart rate were determined.

2.3.1. Electrical stimulation of the perivascular (vasodepressor) sensory outflow

In the first set of rats (n = 75), the pithing rod was replaced by an electrode enamelled except for 1.5 cm length 9 cm from the tip, so that the uncovered segment was situated at T_9-T_{12} of the spinal cord, and an indifferent electrode was placed dorsally (Taguchi et al., 1992; Villalón et al., 2008). Before electrical stimulation, the animals received (i.v.): (i) a bolus injection of gallamine (25 mg/kg) to avoid the electrically induced muscular twitching; and (ii) 10 min later, a continuous infusion of hexamethonium (2 mg/kg.min) to block electrically induced vasopressor responses that are produced by stimulation of the preganglionic sympathetic vasopressor outflow (Villalón et al., 1998). Then, this set of rats was divided into 3 groups (n = 25 each).

The first group received an i.v. continuous infusion of methoxamine (20 µg/kg.min); after 10 min (when diastolic blood pressure was maintained at around 135 mm Hg; see Results section) this group was subdivided into five subgroups that received (i.v.): (i) nothing (control group; n = 5); (ii) an infusion of vehicle (saline, 0.02 ml/min; n = 5); or (iii) infusions of CP-93,129 (0.1, 1 and 10 µg/kg.min; n = 5 for each dose). Twenty minutes later diastolic blood pressure and heart rate were determined again and then the perivascular sensory outflow was electrically stimulated during the above treatments to elicit vasodepressor responses by applying 10-s trains of monophasic, rectangular pulses (2 ms, 50 V), at increasing frequencies of stimulation (0.56, 1, 1.8, 3.1 and 5.6 Hz), as previously described (Villalón et al., 2008). When diastolic blood pressure had returned to baseline levels, the next frequency was applied. This procedure was systematically performed until the stimulus–response curve had been completed.

The second group received an i.v. infusion of methoxamine (20 µg/kg.min) followed, after 10 min, by an infusion of saline (0.02 ml/min). Ten minutes later, this group was subdivided into five subgroups (n = 5 each) comprising i.v. bolus injections of, respectively: (i) saline (1 ml/kg, control group); (ii) dimethylsulfoxide (DMSO) 2.5% (1 ml/kg); (iii) GR127935 (100 µg/kg); (iv) SB224289 (300 µg/kg); and (v) BRL15572 (300 µg/kg). After 10 min, a stimulus-response curve was constructed as described above during the infusion of methoxamine.

The third group received (i.v.) an infusion of methoxamine (20 µg/kg.min) followed, after 10 min, by an infusion of CP-93,129 (1 µg/kg.min). Ten minutes later, this group was subdivided into five subgroups (n = 5 each) comprising i.v. bolus injections of, respectively: (i) saline (1 ml/kg, control group); (ii) DMSO 2.5% (1 ml/kg); (iii) GR127935 (100 µg/kg); (iv) SB224289 (300 µg/kg); and (v) BRL15572 (300 µg/kg). After 10 min, a stimulus–response curve was constructed as described above, during the infusion of CP-93,129 (1 µg/kg.min).

It is important to emphasize that only one stimulus–response curve was carried out per animal since tachyphylaxis of the vasodepressor responses was observed when eliciting a second stimulus–response curve (Villalón et al., 2008; Lozano-Cuenca et al., 2009).

2.3.2. Administration of exogenous α-CGRP

The second set of rats (n = 15) was prepared as described above, but the pithing rod was left throughout the experiment and the administration of both gallamine and hexamethonium was omitted. This set received an i.v. continuous infusion of methoxamine (20 µg/kg.min); after 10 min (when diastolic blood pressure was maintained at around 125 mm Hg; see Results section) this set was divided into three groups ($n=5$ each) that received (i.v.), respectively: (i) nothing (control group); (ii) vehicle (saline; 0.02 ml/min); or (iii) CP-93,129 (1 µg/kg.min). Twenty minutes later, the values of diastolic blood pressure and heart rate were determined again and then the vasodepressor responses elicited by i.v. bolus injections of exogenous α-CGRP (0.1, 0.18, 0.31, 0.56 and 1 µg/kg) were examined in these three groups during the infusions of methoxamine and/or CP-93,129.

2.4. Other procedures applying to all protocols

The doses of hexamethonium, methoxamine and CP-93,129 were continuously infused at a rate of 0.02 ml/min by a WPI model sp100i pump (World Precision Instruments Inc., Sarasota, FL, U.S.A.). The doses of CP-93,129 were selected from preliminary experiments. Moreover, the interval between the different frequencies of stimulation/doses of α-CGRP were dependent on the duration of the resulting vasodepressor responses (5-10 min), as we waited until diastolic blood pressure had returned to baseline values.

2.5. Compounds

Apart from the anaesthetic (diethyl ether), the compounds used in this study (obtained from the sources indicated) were: gallamine triethiodide, hexamethonium chloride, rat α-CGRP, methoxamine hydrochloride, SB224289 (2,3,6,7-tetrahydrospiro-1′-methyl-5- [2′-methyl-4′-(5-methyl-1,2,4-oxadiazol-3-yl)biphenyl-4-carbonyl][furo [2,3-f]indole-3-spiro-4′-piperidine) hydrochloride; BRL15572 [1-(3-chlorophenyl)-4-(3,3-diphenyl(2-(S,R)hydroxy propany) piperazine] hydrochloride (both gifts: Dr. A.A. Parsons, GlaxoSmithKline, Harlow, Essex U.K.), GR127935 (N-[4-methoxy-3-(4-methyl-1-piperazinyl) phenyl]-2′-methyl-4′-(5-methyl-1,2,4-oxadiazol-3-yl)[1,1-biphenyl]-4-carboxamide hydrochloride monohydrate (gift: GlaxoSmithKline, Stevenage, Hertfordshire, U.K.); CP-93,129 (1,4-dihydro-3-(1,2,3,6-tetrahydro-4-pyridinyl)-5H-pyrro l[3,2-b]pyridin-5-one dihydrochloride (gift: Pfizer, Inc., Groton, Conn, U.S.A.); and propylene glycol (J.T. Baker, Mexico City, Mexico). SB224289 and BRL15572 were dissolved in DMSO (2.5 %). This vehicle had no effect on baseline diastolic blood pressure or heart rate (data not shown). Fresh solutions were prepared for each experiment. The doses of antagonists refer to their respective salts, whereas those of the agonists refer to their free base.

2.6. Data presentation and statistical evaluation

All data in the text, tables and figures, unless otherwise stated, are presented as mean ± S.E.M. The peak changes in diastolic blood pressure by electrical stimulation or exogenous α-CGRP were expressed as percent change from baseline. The difference in the absolute values of diastolic blood pressure and heart rate within one subgroup of animals before and during the continuous infusions of methoxamine (20 µg/kg.min) and/or CP-93,129 (0.1, 1 and 10 µg/kg.min) was evaluated with paired Student's t test. Moreover, a one-way analysis of variance was used to compare the absolute values of diastolic blood pressure and heart rate obtained: (i) before and during the continuous infusions of methoxamine (20 µg/kg.min) and/or CP-93,129 (0.1, 1 and 10 µg/kg.min) in the different subgroups; or (ii) during the continuous infusions of methoxamine (20 µg/kg.min) and/or CP-93,129 (1 µg/kg.min) before, immediately after and 10 min after administration of saline, DMSO, GR127935, SB224289 or BRL15572. Finally, the vasodepressor responses induced by electrical stimulation or exogenous α-CGRP in the different subgroups of animals were compared with a two-way analysis of variance (randomized block design). The one- and two-way analysis of variance were followed, if applicable, by the Student–Newman–Keuls' post hoc test. Statistical significance was accepted at $P<0.05$.

3. Results

3.1. Systemic haemodynamic variables

Baseline diastolic blood pressure and heart rate in the 90 pithed rats were 49 ± 4 mm Hg and 253 ± 3 bpm, respectively; these variables remained unchanged after gallamine or hexamethonium (data not shown). Table 2 shows that diastolic blood pressure and heart rate were significantly increased after the infusion of methoxamine. Similar effects were obtained in the animals receiving i.v. bolus injections of α-CGRP (data not shown). It is noteworthy that the absolute values of diastolic blood pressure and heart rate obtained in the different subgroups before, immediately after and 10 min after administration of an i.v. bolus injection of saline, DMSO 2.5% or the antagonists (during the infusion of methoxamine and/or CP-93,129), were not significantly different ($P>0.05$) (Table 3). Except when constructing stimulus-response curves (see below), these effects were sustained throughout the experiments (Fig. 1).

3.2. Vasodepressor responses produced by electrical stimulation or i.v. bolus injections of exogenous α-CGRP

Fig. 2 shows that during methoxamine infusion (control; 20 µg/kg.min), electrical stimulation of the perivascular sensory outflow resulted in frequency-dependent vasodepressor responses that: (i) remained unaltered during the infusion of vehicle (Fig. 2A); and (ii) were smaller during the infusion of CP-93,129 (1 µg/kg.min) (Figs. 1B and 2B). In both cases, the vasodepressor responses appeared about 10 s after starting each electrical stimulus and reached a maximum 1 min after the stimulus had ended. In addition, this inhibition was dose-dependent as 0.1 µg/kg.min CP-93,129 significantly inhibited only the responses induced by 1.8 and 3.1 Hz, whilst 1 and 10 µg/kg.min CP-93,129 inhibited the responses induced by 1.8, 3.1 and 5.6 Hz (there were no significant differences between the responses obtained during these treatments).

On the other hand, Fig. 2C and D shows that during i.v. continuous infusions of methoxamine (control; 20 µg/kg.min), i.v. bolus injections of exogenous α-CGRP elicited dose-dependent vasodepressor responses. These responses remained unchanged during the infusions of: (i) vehicle (0.02 ml/min; Fig. 2C) or (ii) CP-93,129 (1 µg/kg.min; Fig. 2D). Irrespective of methoxamine and/or CP-93,129 being infused, the vasodepressor responses to α-CGRP were immediate and reached a maximum 1 min after the corresponding i.v. injection had been given.

Table 2
Baseline values of diastolic blood pressure and heart rate before and 20 min after administration of an i.v. continuous infusion of methoxamine (20 µg/kg.min), vehicle (saline; 0.02 ml/min) or CP-93,129 (0.1, 1 and 10 µg/kg.min) during a continuous infusion of methoxamine (20 µg/kg.min; control).

Treatment	Dose (µg/kg.min)	n	Diastolic blood pressure (mm Hg)		Heart rate (beats/min)	
			Before	After	Before	After
Methoxamine	20	5	54 ± 2	130 ± 4[a]	251 ± 3	283 ± 4[a]
Vehicle[b]	0.02 ml/min	5	52 ± 2	138 ± 9[a]	245 ± 2	287 ± 5[a]
CP-93,129[b]	0.1	5	48 ± 7	143 ± 9[a]	249 ± 4	270 ± 5[a]
	1	5	48 ± 3	139 ± 11[a]	267 ± 2	296 ± 3[a]
	10	5	47 ± 5	118 ± 4[a]	251 ± 3	279 ± 9[a]

[a] $P<0.05$, after vs before from the corresponding baseline value (paired t test). It is noteworthy that the absolute values of diastolic blood pressure and heart rate obtained in the different subgroups, before and 30 min after the continuous infusions of methoxamine and/or CP-93,129, were not significantly different ($P>0.05$). All values are expressed as mean ± S.E.M.

[b] During a continuous infusion of methoxamine (20 µg/kg.min).

Table 3
Values of diastolic blood pressure and heart rate during the infusion of methoxamine (20 mg/kg.min): (i) before; (ii) immediately after; and (iii) 10 min after i.v. administration of saline, DMSO 2.5% or the antagonists (GR127935, SB224289 or BRL15572).

Treatment	Dose (μg/kg)	*n*	Diastolic blood pressure (mm Hg)			Heart rate (bpm)		
			Before	Immediately after	10 min after	Before	Immediately after	10 min after
Saline	1[a]	5	138 ± 11	143 ± 10	136 ± 10	296 ± 6	293 ± 5	297 ± 6
DMSO 2.5 %	1[a]	5	134 ± 15	145 ± 15	137 ± 17	286 ± 13	267 ± 15	290 ± 16
GR127935	100	5	136 ± 12	149 ± 12	138 ± 10	277 ± 2	275 ± 3	277 ± 2
SB224289	300	5	147 ± 4.6	150 ± 10	134 ± 9	285 ± 6	279 ± 6	287 ± 4
BRL15572	300	5	122 ± 8	113 ± 9	117 ± 6	288 ± 10	286 ± 13	294 ± 12

All values are expressed as mean ± S.E.M. [a]Saline and DMSO 2.5% were given at a dose of 1 ml/kg.

The decreases in diastolic blood pressure returned to baseline levels within 5–10 min after electrical stimulation or α-CGRP as previously reported (Villalón et al., 2008; Lozano-Cuenca et al., 2009). In view that 1 μg/kg.min CP-93,129 inhibited the electrically-induced vasodepressor responses (Fig. 2B), without significantly affecting those by exogenous α-CGRP (Fig. 2D), this dose of CP-93,129 was chosen for further pharmacological analysis.

3.3. Effect of vehicles or antagonists per se on the electrically-induced vasodepressor responses during an infusion of methoxamine

Fig. 3 shows that during methoxamine infusion (20 μg/kg.min), the vasodepressor responses to electrical stimulation in control animals did not significantly differ from those elicited in animals pretreated with an i.v. bolus injection of: (i) saline (1 ml/kg; Fig. 3A); (ii) DMSO 2.5% (1 ml/kg; Fig. 3B); or (iii) the antagonists GR127935 (5-$HT_{1B/1D}$, 100 μg/kg; Fig. 3C), SB224289 (5-HT_{1B}, 300 μg/kg; Fig. 3D) or BRL15572 (5-HT_{1D}, 300 μg/kg; Fig. 3E). The above results indicate that these compounds, at the doses used and under our experimental conditions, were essentially devoid of any effect per se on the electrically induced vasodepressor responses.

3.4. Effect of vehicles or antagonists on CP-93,129-induced inhibition of the electrically induced vasodepressor responses

Fig. 4 shows that CP-93,129 (1 μg/kg.min) induced an inhibition of the electrically-induced vasodepressor responses (as compared to those during 20 μg/kg.min methoxamine alone; control response). These CP-93,129-induced inhibition: (i) remained practically unaltered ($P > 0.05$) in the animals pretreated with 1 ml/kg saline (Fig. 4A) or 1 ml/kg DMSO 2.5% (Fig. 4B); and (ii) was significantly antagonized in animals pretreated with GR127935 (100 μg/kg; Fig. 4C) or SB224289 (300 μg/kg; Fig. 4D), but was not significantly modified in the animals pretreated with BRL15572 (300 μg/kg; Fig. 4E). It must be emphasized that the doses of the above antagonists were high enough to completely block their respective receptors in the cardiovascular system of the rat (Sánchez-López et al., 2004).

4. Discussion

4.1. General

Apart from the implications discussed below, our study shows that the rodent 5-HT_{1B} receptors agonist, CP-93,129 (1 μg/kg.min), inhibited

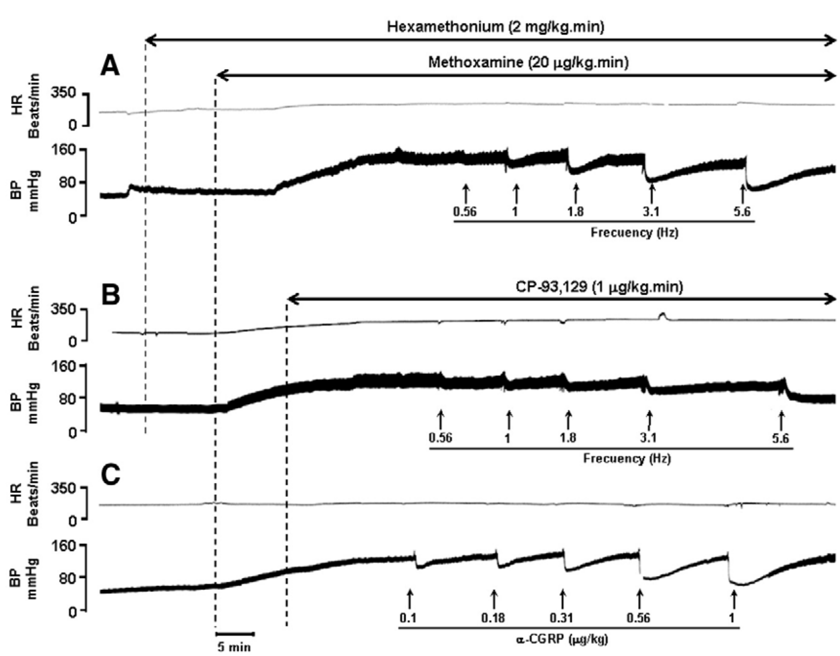

Fig. 1. Original experimental tracings illustrating the vasodepressor responses induced by (A, B) electrical stimulation of the perivascular sensory outflow or (C) i.v. bolus injections of exogenous α-CGRP in pithed rats. Note that during the continuous infusion of methoxamine (20 μg/kg.min) plus CP-93,129 (1 μg/kg.min) only the vasodepressor responses induced by electrical stimulation were attenuated (B), versus control (A), but not during the bolus injections of α-CGRP (C). This finding suggests that the inhibition of the vasodepressor responses by CP-93,129 is prejunctional in nature. BP, blood pressure; HR, heart rate.

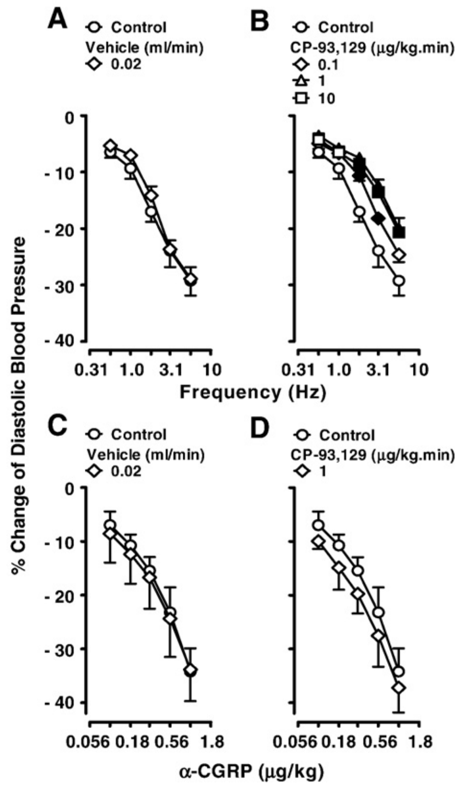

Fig. 2. Vasodepressor responses induced by electrical stimulation (A, B) or i.v. bolus injections of exogenous α-CGRP (C, D) elicited during i.v. continuous infusions of vehicle (saline; 0.02 ml/min) or CP-93,129 (0.1, 1 and 10 µg/kg.min) ($n = 5$ each dose). For the sake of clarity, empty symbols depict either control (methoxamine; 20 µg/kg.min) responses (○) or non-significant responses (◇ △ □) vs. control. Solid symbols (◆ ▲ ■) represent significantly different responses ($P < 0.05$) vs. control.

the vasodepressor responses induced by stimulation of the perivascular (CGRPergic) sensory outflow, but not those by exogenous α-CGRP. Hence such inhibition is mediated by prejunctional receptors which, upon activation, inhibit the vasodepressor sensory outflow. The pharmacological profile of these receptors correlates with the 5-HT_{1B} receptor subtype as CP-93,129-induced inhibition was blocked by GR127935 or SB224289, but not by BRL15572.

Regarding our experimental protocols, it is important to note that only one stimulus–response curve could be carried out per animal since tachyphylaxis was observed when eliciting a second stimulus–response curve (Villalón et al., 2008). This phenomenon may involve depletion of neuronal CGRP, uncoupling the G protein, sequestration and/or receptor down-regulation (see Buxton, 2006). However, our experimental model did not allow us to assess the extent to which each of these mechanisms is involved. In addition, it must be emphasized that we did not measure sensorial nerve activity directly, but the electrically-induced neuropeptide release (i.e. CGRP) in the systemic vasculature could be estimated indirectly by the measurement of the evoked vasodepressor responses, as shown by Taguchi et al. (1992).

4.2. Systemic haemodynamic effects produced by the different treatments

As previously reported (Villalón et al., 2008), the sustained increase in diastolic blood pressure by methoxamine (see Table 2) results from an increase in peripheral vascular resistance (see Westfall and Westfall, 2006), and this effect is mainly mediated by stimulation of vascular α_1-adrenoceptors (Decker et al., 1984; Hoffman, 2001). In contrast, we have no clear-cut explanation for the increase in heart rate induced by the infusion of methoxamine (see Section 3.1), which does not activate β-adrenoceptors (see Westfall and Westfall, 2006), although we cannot categorically rule out the activation of cardiac α_1-adrenoceptors (Terzic et al., 1993).

Furthermore, the fact that the infusions of CP-93,129 did not significantly modify the increase in diastolic blood pressure during methoxamine infusion (see Table 2) may be explained in terms that activation of peripheral 5-HT_{1B} receptors: (i) does not exert significant effects on blood pressure, as previously reported (De Vries et al., 1997); and (ii) is involved in sumatriptan-induced contractions of carotid, cranial and coronary arteries, responses antagonized by SB224289, but not BRL15572 (De Vries et al., 1998; Maassen Vandenbrink et al., 2000; Verheggen et al, 1998; Wackenfors et al., 2005).

In addition, as previously reported (Taguchi et al., 1992; Villalón et al., 2008), electrical stimulation of the perivascular sensory outflow produces vasodepressor responses only after continuous i.v. infusions of hexamethonium (to block autonomic outflow) and methoxamine (for a sustained increase in blood pressure). Since these vasodepressor responses remained unaltered during the continuous infusion of saline (Fig. 2A) as well as after the i.v. bolus injections of saline (Fig. 3A) or DMSO 2.5% (Fig. 3B), the effects induced by the compounds cannot be attributed to these vehicles.

Most importantly, the fact that GR127935, SB224289 or BRL15572 had no effects on baseline diastolic blood pressure and heart rate (Table 3) or on the electrically-induced vasodepressor responses (Fig. 3)

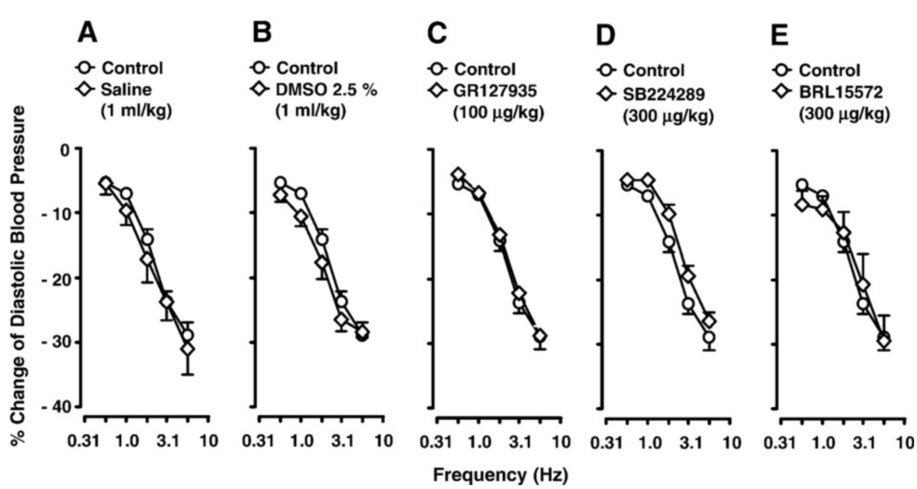

Fig. 3. Effect of i.v. bolus injections of saline (A; 1 ml/kg), DMSO 2.5% (B; 1 ml/kg), GR127935 (C; 100 µg/kg), SB224289 (D; 300 µg/kg) or BRL15572 (E; 300 µg/kg) per se ($n = 5$ each dose) on the electrically induced vasodepressor responses induced during an i.v. continuous infusion of methoxamine (20 µg/kg.min) ($n = 5$ for each group).

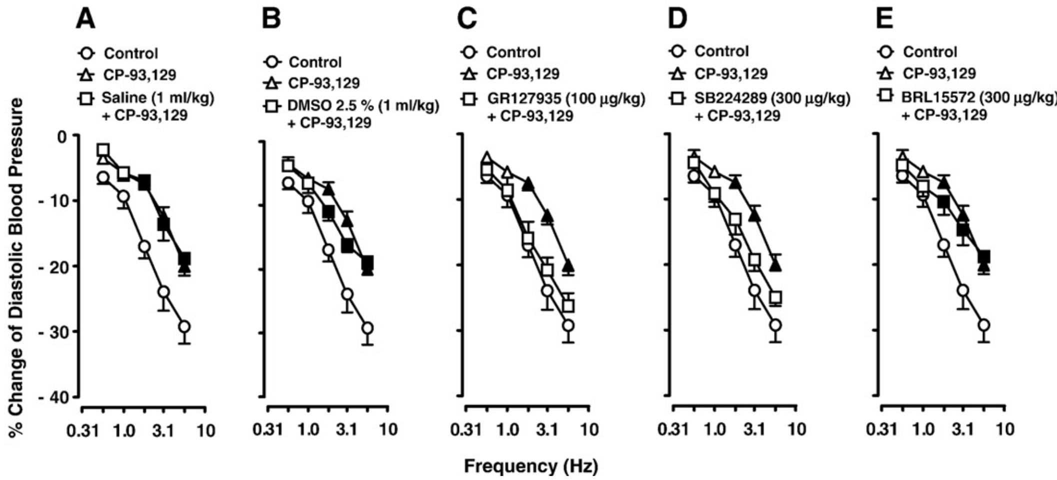

Fig. 4. Effect of i.v. bolus injections of saline (A; 1 ml/kg), DMSO 2.5% (B; 1 ml/kg) GR127935 (C; 100 µg/kg), SB224289 (D; 100 µg/kg) or BRL15572 (E; 100 µg/kg) on the inhibition induced by CP-93,129 (1 µg/kg.min) of the electrically induced vasodepressor responses. The control group represents that of animals receiving an i.v. continuous infusion of methoxamine (20 µg/kg.min), which is shown for comparison. For the sake of clarity, empty symbols depict either control (methoxamine; 20 µg/kg.min) responses (○) or non-significant responses (△ □) vs. control. Solid symbols (▲ ■) represent significantly different responses (*P*<0.05) vs. control.

suggests that: (i) these compounds, at the doses used, are devoid of any effect per se on the neurovascular CGRPergic transmission; and (ii) any effect of a given antagonist on CP-93,129-induced inhibition of the vasodepressor sensory outflow is attributable to a direct interaction of the antagonist with its respective receptor on the perivascular sensory nerves, rather than to changes in the vasodepressor responses and/or in the baseline systemic haemodynamic values.

4.3. The role of prejunctional receptors in inhibiting the vasodepressor sensory outflow: correlation with the 5-HT$_{1B}$ subtype

Since 1 µg/kg.min CP-93,129 inhibited the vasodepressor responses to electrical stimulation without affecting those to i.v. bolus injections of α-CGRP (Fig. 2), it may be inferred that the inhibition to CP-93,129 could be mediated by a prejunctional inhibitory action on the perivascular sensory nerves of the systemic vasculature. This, in turn, may lead to a decrease in CGRP release, as previously described for clonidine in the same experimental model (Villalón et al., 2008). The pharmacological profile of the receptors involved in this CP-93,129-induced inhibition closely resembles the 5-HT$_{1B}$ receptor subtype on the basis that: (i) CP-93,129 displays a much higher affinity for rat 5-HT$_{1B}$ than for 5-HT$_{1A}$ and 5-HT$_{1D}$ receptors (see Table 1); and (ii) CP-93,129-induced inhibition was blocked by SB224289 (a selective 5-HT$_{1B}$ receptor antagonist) or GR127935 (a 5-HT$_{1B/1D}$ receptor antagonist), but not by BRL15572 (a selective 5-HT$_{1D}$ receptor antagonist) (see Table 1) at doses high enough to completely block prejunctional 5-HT$_{1B}$ and 5-HT$_{1D}$ receptors in the rat (Sánchez-López et al., 2004). Admittedly, the affinities of SB224289 and BRL15572 for, respectively, 5-HT$_{1B}$ and 5-HT$_{1D}$ receptors in the rat have not been published, but functional studies of prejunctional sympatho-inhibitory 5-HT$_{1B}$ and 5-HT$_{1D}$ receptors demonstrate the blocking properties of these compounds in the rat (Sánchez-López et al., 2004). Although GR127935 displays a high affinity for 5-HT$_{1D}$ receptors (pK_i = 8.6) and a moderate affinity for 5-HT$_{1A}$ receptors (pK_i = 7.2) (Table 1), the involvement of these receptors seems unlikely for the reasons described above, and because additionally: (i) SB224289 has a low affinity for 5-HT$_{1A}$ receptors (Table 1); and (ii) BRL15572 displays a similar affinity for 5-HT$_{1A}$ and 5-HT$_{1D}$ receptors. Obviously, our suggestion supporting the role of CP-93,129-sensitive 5-HT$_{1B}$ receptors is based on the assumption that species differences between the binding of SB224289 and BRL15572 to rodent and human 5-HT$_{1B}$ and 5-HT$_{1D}$ receptors do not play a major role (see Table 1).

4.4. Transductional evidence and possible locus of the inhibitory 5-HT$_{1B}$ receptors

Admittedly, we have no direct evidence that the inhibition produced by CP-93,129 in our experiments involves inhibition of adenylyl cyclase, inactivation of Ca^{2+} channels and/or activation of inwardly rectifying K^+ channels in sensory neurons. Nevertheless, all 5-HT$_1$ receptor subtypes are, by definition, coupled to $G_{i/o}$ proteins that inhibit adenylyl cyclase activity, inactivate Ca^{2+} channels and/or activate inwardly rectifying K^+ channels (Hoyer et al., 1994; Hannon and Hoyer, 2008). These are signal transduction systems usually associated with inhibition of neurotransmitters release (Rand et al., 1987; Boehm and Kubista, 2002; De Jong and Vehage, 2009). More specifically, CP-93,129 inhibited forskolin-induced adenylate cyclase activity and this is directly mediated by 5-HT$_{1B}$ receptors (Macor et al., 1990).

On the other hand, one might speculate upon the possible locus of the 5-HT$_{1B}$ receptors that inhibit the vasodepressor sensory outflow. In this respect: (i) the presence of 5-HT$_{1B}$ receptor *m*RNA has been detected in rat DRG neurons (Pierce et al., 1996; Nicholson et al., 2003); and (ii) 5-HT$_{1B}$ receptors are the predominant type labeled by [^{3}H]5-HT in cultured DRG neurons, representing 60% of the specific [^{3}H]5-HT binding sites (Chen et al., 1998). Central mechanisms can be entirely excluded since pithed rats were used.

4.5. Final considerations on our experimental conditions and the use of 5-HT as an agonist

It must be highlighted that, under our experimental conditions, 5-HT (the endogenous ligand) is not an appropriate agonist to identify specific 5-HT$_1$ receptor subtypes; it can stimulate all 5-HT receptors, including 5-HT$_7$ receptors (producing vasodepressor responses) and all 5-HT$_1$ receptor subtypes (most of which produce prejunctional inhibition of neurotransmitter release) (Villalón and Centurión, 2007). In keeping with these complex actions, i.v. continuous infusions of 5-HT (instead of CP-93,129) in our experiments produced a significant and sustainable decrease in diastolic blood pressure (data not shown); this hypotensive action represents an impediment for us to determine the inhibition produced by 5-HT on the vasodepressor sensory outflow, as diastolic blood pressure must be maintained at around 135 mm Hg, as previously reported (Villalón et al., 2008). Hence, only selective agonists for 5-HT receptors devoid of any hypotensive action (as CP-93,129) can be used in our experiments.

5. Conclusion

The above results, taken together, suggest that activation of prejunctional 5-HT_{1B} receptors inhibits the vasodepressor sensory CGRPergic outflow in pithed rats.

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